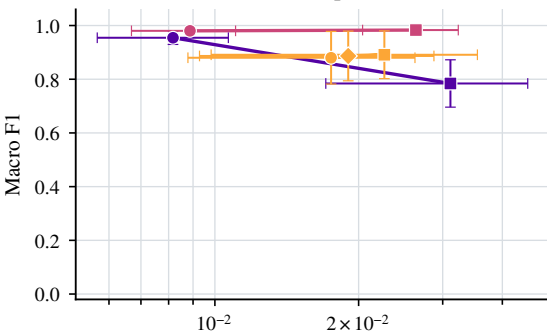
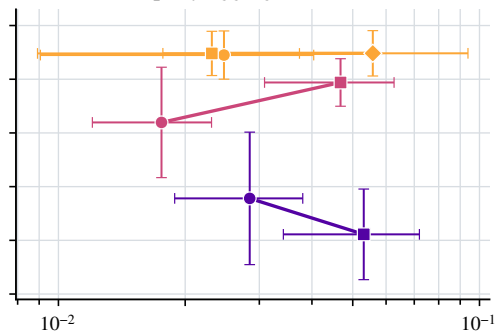


—●— LLM
 —■— CodeAct
 —◆— RLM
 ● 100
 ■ 500
 ◆ Full

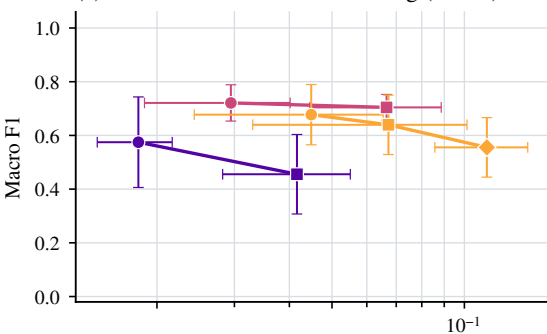
(a) Tier 1: Structural lookup ($n = 10$)



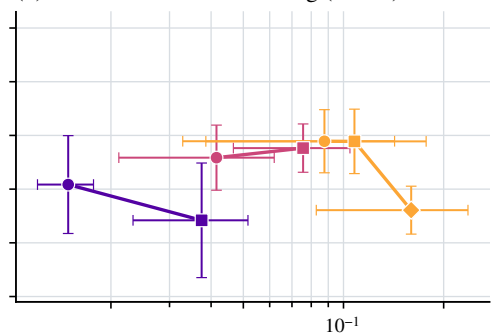
(b) Tier 2: Property aggregation ($n = 20$)



(c) Tier 3: Reaction-level reasoning ($n = 35$)



(d) Tier 4: Relational reasoning ($n = 35$)



Recorded cost (USD) / trajectory

Recorded cost (USD) / trajectory